

CAR PARKING SLOT DETECTION USING YOLO ALGORITHM

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Abstract - Finding free slots in the parking area has become a major problem that the modern society is facing nowadays, which also leads to traffic overcrowding at the parking area. As the number of vehicles are increasing at a quick pace find parking slots has become difficult. The proposed paper helps to solve the parking problem. In traditional parking system a person has to find vacant slot or a gate keeper needs to inform about the available parking slot at parking area. To address the issue, a system where the user will be able to know the number of vacant slots in a particular parking area along with total number of slots at the entrance of the parking area. As a critical component of this project, by built a car parking slot detection system which collects data from a camera at parking area. Today's most popular object detection model YOLO, is trained to achieve desired results with good precision scores at real-time speed. The model will be getting all the parking slots that are present in the parking area, to which marking boxes are drawn around for training. Real time processing is performed on the obtained data by the model to find whether the slot is vacant or occupied. Thus, it gives the information of available vacant slots to park the car at parking area.

Key Words: YOLO, Object detection, Computer vision, Machine learning.

I. INTRODUCTION

Over the past decade, the number of vehicles has soared significantly, straining the existing transportation infrastructure and parking facilities. This has led to widespread traffic congestion and parking shortages. To address these issues, real-time detection of available parking spaces is crucial. By providing drivers with up-to-date information about empty stalls, it can reduce traffic buildup, enhance the efficiency of parking lot operations, and minimize the time spent searching for an empty space. Considerable

efforts have been made to tackle the transportation challenges posed by the surge in vehicles. To address parking challenges, various technologies like sensor-based computer vision classifiers have emerged. However, their high cost and limitations in efficiency and robustness present obstacles. To resolve these issues, researchers have devised parking lot recognition techniques utilizing resources like video cameras and sensors to identify occupied and unoccupied parking spaces. One such approach leverages YOLO (You Only Look Once), an object detection algorithm powered by Convolutional Neural Networks (CNN). This project aims to address parking space limitations through video footage from surveillance cameras. By analyzing these videos, the system can accurately identify vacant parking spaces, making it easier for drivers to locate available spots. This solution eliminates the time-consuming process of searching for parking, ensuring that drivers can park quickly and efficiently.

II. LITERATURE SURVEY

In [1] an automated vehicle parking slot detection system utilizes Mask R CNN and PSDL for efficient instance segmentation, marking point detection, and free space identification between vehicles. The approach focuses on vision-based car parking technology using web cameras and addresses infrastructure maintenance and cost concerns associated with sensor-based systems. Successful implementation demonstrates high accuracy in object detection, emphasizing practical implications for real-world applications.

In [2] a car detection-based algorithm for automatic parking space detection aims to automate mapping by emphasizing car detection and available space identification. It indicates potential for a 90% reduction in human effort and nearly 100% detection when all spots are occupied. The approach's adaptability across varied parking lots and camera angles is attributed to its focus on car features rather than parking lot features.

In [3] a method for small target detection utilizes dilated modules, feature fusion, and passthrough modules. It